

Assignment No. 4

Client.java

```
import java.io.*;
import java.net.*;

public class Client
{
    public static void main(String args[]) throws Exception
    {
        Socket s = new Socket("localhost", 6666);

        DataOutputStream dos = new DataOutputStream(s.getOutputStream());
        DataInputStream dis = new DataInputStream(s.getInputStream());

        long clientTime = System.currentTimeMillis();

        System.out.println("Client Time: " + clientTime);

        dos.writeLong(clientTime);

        long adjustment = dis.readLong();

        long updatedTime = clientTime + adjustment;

        System.out.println("Adjusted Time: " + updatedTime);

        s.close();
    }
}
```

Server.java

```
import java.io.*;
```

```
import java.net.*;
import java.util.*;

public class Server
{
    public static void main(String args[]) throws Exception
    {
        ServerSocket ss = new ServerSocket(6666);
        System.out.println("Server started...");

        Socket s1 = ss.accept();
        Socket s2 = ss.accept();

        DataInputStream dis1 = new DataInputStream(s1.getInputStream());
        DataInputStream dis2 = new DataInputStream(s2.getInputStream());

        DataOutputStream dos1 = new DataOutputStream(s1.getOutputStream());
        DataOutputStream dos2 = new DataOutputStream(s2.getOutputStream());

        long t1 = dis1.readLong();
        long t2 = dis2.readLong();

        long serverTime = System.currentTimeMillis();

        System.out.println("Client1 Time: " + t1);
        System.out.println("Client2 Time: " + t2);
        System.out.println("Server Time: " + serverTime);

        long avg = (t1 + t2 + serverTime) / 3;

        dos1.writeLong(avg - t1);
```

```
dos2.writeLong(avg - t2);
```

```
System.out.println("Average Time: " + avg);
```

```
s1.close();
```

```
s2.close();
```

```
ss.close();
```

```
}
```

```
}
```

Output

Server:

```
C:\Users\anuja\OneDrive\Documents\DS_prac>cd Assignment4
```

```
C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment4>notepad Server.java
```

```
C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment4>notepad Client.java
```

```
C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment4>javac *.java
```

```
C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment4>java Server
```

Server started...

Client1 Time: 1777002075240

Client2 Time: 1777002134908

Server Time: 1777002134909

Average Time: 1777002115019

Client 1 :

```
C:\Users\anuja\OneDrive\Documents\DS_prac>cd Assignment4
```

```
C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment4>java Client
```

Client Time: 1777002075240

Adjusted Time: 1777002115019

Client2:

C:\Users\anuja\OneDrive\Documents\DS_prac>cd Assignment4

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment4>java Client

Client Time: 1777002134908

Adjusted Time: 1777002115019